

Sriram Bharadwaj

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🔗 [Google Scholar](#)

Education

University of California, Los Angeles

PhD in Theoretical Physics

Sept 2021 – 2027

(anticipated)

- Actively on the job market for postdoctoral positions in 2026-27
- **PhD advisors:** Eric D'Hoker and Leo Zhou, **Mentors:** Di Luo and Zhong-Bo Kang
- **Awarded**
 1. “UCLA Physics & Astronomy Graduate Research Award” (2026-27), amount: \$30k
 2. “UCLA Robert Finkelstein Fellowship in Theoretical Physics” (2022), amount: \$6k
 3. “Mani L. Bhaumik Graduate Fellowship in Theoretical and Quantum Physics” (2023-26), amount: \$6k/summer
- Semi-finalist (top 20 of 133) in the XPrize Quantum Applications competition (2025):
“CV-QAOA: Efficient Low-Depth Quantum Optimization for Continuous Variables” – with Di Luo and Leo Zhou

University of Oxford

MSc in Mathematical and Theoretical Physics

Oct 2020 – Sept 2021

- Graduated with distinction and completed a double-unit master's dissertation on string amplitudes

University of California, Los Angeles

BS in Mathematics and Applied Science

Sept 2017 – Sept 2020

- Graduated *summa cum laude* and was on Dean's honors list every quarter

Publications (in reverse chronological order)

1. **(To appear) Confinement Versus Screening in the AdS_2 Schwinger Model from Bosonization and Tensor Networks** *July 2026 (anticipated)*
with Zhong-Bo Kang and Jack Isen [View Draft](#)
2. **(To appear) CV-QAOA: Efficient Low-Depth Quantum Optimization of Continuous Variables** *July 2026 (anticipated)*
with Di Luo and Leo Zhou, [View Draft](#)
3. **Information-Theoretic Scaling Laws of Neural Quantum States** *Mar 2026*
with Y. Lu, D. Rathore, D. Luo [arXiv:2603.23468](#)
4. **Hamiltonian Formulation of Lattice QED₃ with One and Two Flavors of Wilson Fermions: Topological Structure and Response** *Mar 2026*
with Di Luo, Lena Funcke, Karl Jansen, et al, [arXiv: 2603.05616](#)
5. **A Path to Quantum Simulations of Topological Phases: (2+1)D Quantum Electrodynamics with Wilson Fermions** *Apr 2025*
with Di Luo, Lena Funcke, Karl Jansen, et al, [arXiv: 2504.21828](#)
6. **Non-Invertible Defects on the Worldsheet** *Sep 2024*
with P. Niro and K. Roumpedakis, published in *JHEP* 03 (2025) 164
7. **Approaching Argyres-Douglas Theories** *Jun 2024*
with Eric D'Hoker, published in *JHEP* 06 (2024) 082

Selected ongoing research (in order of planned submission)

Exponentially Accurate Adiabatic State Preparation for Lattice Gauge Theories and Bosonic Systems

with Di Luo

- Proves n th-order adiabatic theorem for truncated unbounded Hamiltonians with errors exponentially small in run-time; applied to $(2+1)\text{D}$ $\text{U}(1)$ and $\text{SU}(2)$ lattice gauge theories and the Hubbard-Holstein model
- Cubic improvement over previously considered algorithms for $(2+1)\text{D}$ lattice gauge theories, exponential improvement over Quantum Monte Carlo/super-polynomial improvement over DMRG for the Hubbard-Holstein model

- VQE to prepare topological ground states of QED₃ with rigorous gate-complexity estimates
- Experimentally validates the topological phase diagram of [2504.21828](#)  and [2603.05616](#)  on superconducting quantum hardware

List of presentations (in reverse chronological order)

- **Seminar:** “Confinement Vs. Screening in the AdS₂ Schwinger Model from Bosonization and Tensor Networks”, Center for Frontiers in Nuclear Science, Stony Brook University, Host: Felix Ringer (Aug 2026)
- **Seminar:** “Quantum Simulations of Topological Phases: (2+1)D Lattice QED with Wilson Fermions”, Boston College, Host: Xiao Chen (Aug 2026)
- **Contributed talk:** “Quantum Simulations of Topological Phases: (2+1)D Lattice QED with Wilson Fermions” (arXiv: [2603.05616](#) ) , IAIFI Workshop, MIT (Aug 2026).
- **Contributed talk:** “Quantum Simulations of Topological Phases: (2+1)D Lattice QED with Wilson Fermions” (arXiv: [2603.05616](#) ) , The 43rd International Symposium on Lattice Field Theory (Lattice 2026), University of Maryland, College Park (Jul 2026).
- **Poster:** “Quantum Simulations of Topological Phases in (2+1)D Lattice Gauge Theory” (arXiv: [2504.21828](#) ) , “QuantHEP 2025”, Lawrence Berkeley National Laboratory (Oct 2025).
- **Contributed talk:** “Quantum Simulations of Topological Phases in (2+1)D Lattice Gauge Theory” (arXiv: [2504.21828](#) ) , NSF Institute for Artificial Intelligence and Fundamental Interactions (IAIFI) Summer School, Harvard (Aug 2025).
- **Contributed talk:** “Probing the frontiers of nuclear physics with AI at the EIC (II)”, Stony Brook (Mar 2025).
- **Contributed talk:** “Non-invertible symmetries on the worldsheet” based on [JHEP 03 \(2025\) 164](#) , “2024 IHES Summer School - Symmetries and Anomalies: a Modern Take”, IHES, France (June 2024).
- Pedagogical presentations in the High-Energy Theory Journal Club at UCLA.
 1. Confinement vs. screening in QED₂ on AdS₂ (2026)
 2. Anomalies and instantons in gauge theory (2024)
 3. Liouville field theory and 2D quantum gravity (2023)
 4. Quantum information theory and entanglement entropy in gravity (2023)
 5. Aspects of confining strings in gauge theory (2022)

List of conferences and schools (in reverse chronological order)

- The 43rd International Symposium on Lattice Field Theory (Lattice 2026), UMD, College Park, 2026
- Institute for AI and Fundamental Interactions Workshop 2026, MIT
- Institute for AI and Fundamental Interactions Summer School 2026, University of Massachusetts, Boston
- Quantum Winter School 2026: Quantum Simulation at Institute for Pure and Applied Mathematics, UCLA
- Institute for AI and Fundamental Interactions Summer School and Workshop 2025, Harvard University
- 2024 Summer School on Symmetries and Anomalies, IHES, France
- 50 years of QCD, UCLA

References

- Eric D’Hoker, Distinguished Research Professor in Theoretical Physics, UCLA,  dhoker@physics.ucla.edu
- Di Luo, Associate Professor of Physics, Tsinghua (formerly Prof. at UCLA ECE),  diluo@tsinghua.edu.cn
- Zhong-Bo Kang, Professor of Physics, UCLA,  zkang@physics.ucla.edu
- Leo Zhou, Assistant Professor of Electrical and Computer Engineering, UCLA,  leoxzhou@ucla.edu

Teaching and Professional Experience

- Refereed a paper submitted to the Quantum Information Processing (QIP) conference, 2025
- Was a teaching assistant for 10+ undergraduate courses, both elementary and advanced
- Appointed as lead teaching assistant to manage and update a course with over 900 students